Tech Bite #2: Propagation -- Skip, Scatter, and the Radio Horizon

How a signal gets far -- line of sight, HF skip off the ionosphere, and the surprise long-distance modes on VHF and UHF.

Group T3C · 11 pool questions · 5-10 minute presentation · Practice exams at eham.net/exams

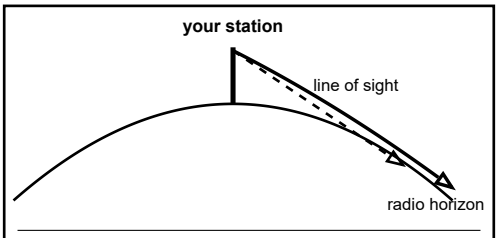
The Big Picture

Bite 1 covered what a radio wave is and what happens to it close to you. This Bite answers a bigger question: how does a signal get **far** – sometimes across town, sometimes across an ocean? The answer depends almost entirely on what the signal does when it heads upward, and on which band you are using. This is where ham radio gets exciting.

Line of Sight and the Radio Horizon

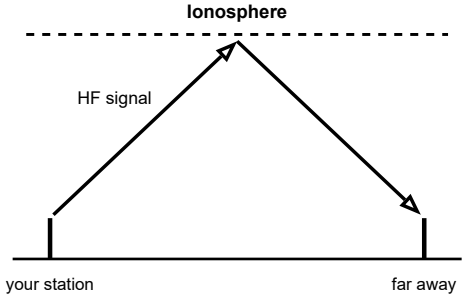
VHF and UHF signals – 2 meters, 70 centimeters, and up – travel mostly in straight lines, called **line of sight**. They reach the horizon and not much past it, because these signals are usually **not** reflected by the ionosphere; they simply keep going up into space. That is why a simplex UHF signal is rarely heard beyond its radio horizon. (T3C01)

One small bonus: the **radio** horizon is a little farther than the **visual** horizon. The atmosphere bends – refracts – radio waves slightly, so they curve a bit past the point where you could see. (T3C11)



VHF/UHF go line of sight. The radio horizon reaches a little past the visual horizon because air bends the waves.

Talking Far: HF Skip off the Ionosphere



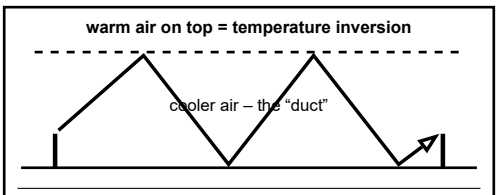
HF skip: a signal goes up, reflects off the ionosphere, and returns to Earth hundreds or thousands of miles away.

Drop down to the HF bands (3-30 MHz) and the rules change completely. The **ionosphere** – that high, electrically charged layer – can **reflect HF signals back down to Earth**. A signal goes up at an angle, bounces, and comes down far away. This is called **skip**, and long-distance contacts this way are far more common on HF than on VHF and higher. (T3C02)

For a Technician, the **10-meter band** is your HF skip window. It works best from **dawn until shortly after sunset, during high sunspot activity** – sunspots energize the ionosphere. (T3C09) At the **peak of the sunspot cycle**, both 6 and 10 meters can open for long-distance F-region contacts. (T3C10)

Surprise Long-Distance on VHF and UHF

Even VHF and UHF can occasionally reach far, through a handful of special effects. You do not have to make these happen – you just need to recognize them when the exam describes them. The most regular one is **tropospheric ducting**, shown at right: a warm layer of air over cooler air traps signals and carries them along.



Tropospheric ducting: a temperature inversion traps VHF/UHF signals and carries them past the horizon, often 300 miles.

Effect	What it does	Q
Sporadic E	Patches of a lower ionosphere layer briefly reflect signals – occasional strong contacts on 10, 6, and 2 meters from beyond the horizon.	T3C04
Tropospheric ducting	A **temperature inversion** (warm air over cooler air) traps VHF/UHF signals, regularly carrying them about 300 miles.	T3C06, T3C08
Meteor scatter	Bounce short bursts off the ionized trail a meteor leaves behind. Works best on **6 meters**.	T3C07
Auroral backscatter	VHF signals bounced off the aurora. They return distorted, with a characteristic **raspy sound**.	T3C03
Knife-edge diffraction	A sharp obstruction, like a mountain ridge, bends signals over its edge to reach stations beyond it.	T3C05

Mental Cheat Sheet

- VHF/UHF travel **line of sight** – rarely past the radio horizon.
- The radio horizon is a bit **farther** than the visual horizon (air bends the waves).
- HF reflects off the **ionosphere** – "skip" – so long distance is common on HF.
- 10 m skip is best **dawn to sunset, high sunspots**; 6 and 10 m open at the sunspot peak.
- **Sporadic E**: surprise strong signals on 10, 6, 2 m.
- **Tropospheric ducting**: about 300 mi on VHF/UHF, from temperature inversions.
- **Meteor scatter**: best on 6 m. **Auroral**: raspy, distorted sound.

Common Traps

- UHF rarely goes far because the ionosphere does **not** reflect it – not because it is "too weak."
- Tropospheric ducting comes from **temperature inversions** – not storms, not sunspots.
- Meteor scatter is best on **6 meters** – not 2 m, not 70 cm.
- The radio horizon is farther because air **bends** the waves – radio does not travel faster than light.
- HF long-distance is the **ionosphere** (skip) – not "smaller antennas" or "less static."

Takeaway: VHF and UHF go line of sight, reaching just past the visible horizon. HF **skips** off the ionosphere and can travel the world – best on 10 meters in the daytime when sunspots are high. And every so often VHF and UHF surprise you, with sporadic E, tropospheric ducting, or meteor scatter carrying signals far beyond the horizon. Knowing which effect does what is most of this group.

Bite 2 Question Bank (T3C)

T3C01. Why are simplex UHF signals rarely heard beyond their radio horizon?
A. They are too weak to go very far
B. FCC regulations prohibit them from going more than 50 miles
C. UHF signals are usually not propagated by the ionosphere
D. UHF signals are absorbed by the ionospheric D region

T3C02. What is a characteristic of HF communication compared with communications on VHF and higher frequencies?
A. HF antennas are generally smaller
B. HF accommodates wider bandwidth signals
C. Long-distance ionospheric propagation is far more common on HF
D. There is less atmospheric interference (static) on HF

T3C03. What is one characteristic of VHF signals received via auroral backscatter?
A. They are often received from 10,000 miles or more
B. They are distorted, with a characteristic raspy sound
C. They occur only during winter nighttime hours
D. They are generally strongest when your antenna is aimed west

T3C04. Which of the following types of propagation is most commonly associated with occasional strong signals on the 10-, 6-, and 2-meter bands from beyond the radio horizon?
A. Backscatter
B. Sporadic E
C. D region absorption
D. Gray-line propagation

T3C05. Which of the following effects may allow radio signals to travel beyond obstructions between the transmitting and receiving stations?
A. Knife-edge diffraction
B. Faraday rotation
C. Quantum tunneling
D. Doppler shift

T3C06. What type of propagation is responsible for allowing over-the-horizon VHF and UHF communications to ranges of approximately 300 miles on a regular basis?
A. Tropospheric ducting
B. D region refraction
C. F2 region refraction
D. Faraday rotation

T3C07. What band is best suited for communicating via meteor scatter?
A. 33 centimeters
B. 6 meters
C. 2 meters
D. 70 centimeters

T3C08. What causes tropospheric ducting?
A. Discharges of lightning during electrical storms
B. Sunspots and solar flares
C. Updrafts from hurricanes and tornadoes
D. Temperature inversions in the atmosphere

T3C09. What is generally the best time for long-distance 10-meter band propagation via the F region?
A. From dawn to shortly after sunset during periods of high sunspot activity
B. From shortly after sunset to dawn during periods of high sunspot activity
C. From dawn to shortly after sunset during periods of low sunspot activity
D. From shortly after sunset to dawn during periods of low sunspot activity

T3C10. Which of the following bands may provide long-distance communications via the ionosphere's F region during the peak of the sunspot cycle?
A. 6 and 10 meters
B. 23 centimeters
C. 70 centimeters and 1.25 meters
D. All these choices are correct

T3C11. Why is the radio horizon for VHF and UHF signals more distant than the visual horizon?
A. Radio signals move somewhat faster than the speed of light
B. Radio waves are not blocked by dust particles
C. The atmosphere refracts radio waves slightly
D. Radio waves are blocked by dust particles

Answer Key

T3C01: C	T3C02: C	T3C03: B	T3C04: B
T3C05: A	T3C06: A	T3C07: B	T3C08: D
T3C09: A	T3C10: A	T3C11: C	